FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Infection Prevention & Control in the Clinical Setting

Clinical skill | Week 2 · Labour, delivery & the puerperium | Sr Kiggundu, Dr Kaduyu

Infection prevention and control (IPC) is the set of everyday practices that stop patients and health workers infecting each other. These are hand hygiene at the 5 moments, PPE chosen by risk, the decontamination-cleaning-HLD/sterilisation chain for instruments, colour-coded waste segregation, sharps safety, and a fast response to exposures, including HIV PEP within 72 hours.

Must know before the ward round

1. Standard precautions apply to EVERY patient every time, whatever the diagnosis. They are hand hygiene, PPE by risk assessment, safe sharps, respiratory hygiene, environmental cleaning, safe linen and waste, instrument reprocessing and safe injections.
2. The WHO 5 Moments are (1) before touching a patient, (2) before a clean/aseptic procedure, (3) after body-fluid exposure risk, (4) after touching a patient, and (5) after touching patient surroundings. Alcohol rub takes 20–30 seconds; soap and water takes 40–60 seconds (use it when hands are visibly soiled).
3. Instrument processing runs decontamination in 0.5% chlorine for 10 minutes, then cleaning with detergent and water, then HLD (boiling 20 min, or 0.1% chlorine or 2% glutaraldehyde 20 min) or sterilisation (autoclave 121 °C at 106 kPa for 20 min unwrapped or 30 min wrapped; dry heat 170 °C for 60 min).
4. Spaulding classification. Critical items (enter sterile tissue) need sterilisation. Semi-critical items (touch mucous membranes, e.g. specula) need at least HLD. Non-critical items (touch intact skin) need cleaning ± low-level disinfection.
5. Uganda waste colour codes are black for non-infectious waste, yellow for infectious waste, red for highly infectious waste, brown for pharmaceutical/chemical waste, and green for non-infectious plastics (per MoH). Sharps go in a yellow safety box, closed at three-quarters full.
6. Needle-stick first aid means letting it bleed without squeezing, washing with soap and running water, and reporting it. HIV PEP is TDF/3TC/DTG (TLD) one tablet daily for 28 days, started as soon as possible (ideally within 2 hours, never after 72 hours). Also assess hepatitis B vaccination status.
7. Transmission risk after a needle-stick from an infected source is about 0.3% for HIV, about 1.8% for hepatitis C, and up to 30% for hepatitis B (HBeAg-positive). Hepatitis B is the biggest occupational threat, and it is vaccine-preventable.
8. Donning order is gown, mask, eye protection, gloves. Doffing order is gloves, eye protection, gown, mask (or gown and gloves together), with hand hygiene after every step where hands may be contaminated.
9. PEP is needed only for a significant exposure (percutaneous, mucous membrane or non-intact skin, with blood or an infectious fluid such as liquor or vaginal secretions) from a source who is HIV-positive or of unknown status. Intact skin needs no PEP. Never delay PEP for the source result.
10. Exposure workup. First aid, report, note the time, rapid HIV test for the worker (positive means ART, not PEP) and the source (with consent), HBsAg, anti-HBs if the response is unknown (10 mIU/mL or more is protective), and a pregnancy test if relevant.
11. An outbreak is more cases than expected in a place and time. Notify the IPC focal person, write a case definition, make a line list, culture before antibiotics, audit practice (antibiotic timing, autoclave log and indicators, chlorine freshness, hand rub, multi-dose vials), control and feed back.
12. Take pus and blood cultures before antibiotics. Ugandan SSI isolates are often ESBL Klebsiella or E. coli and MRSA, and the same organism in several patients points to a common source.

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1. Purpose and Indications

Infection prevention and control (IPC) means the evidence-based practices that prevent **health-care-associated infections (HAIs)** in patients and **occupational infections** in health workers. On an Ob/Gyn rotation it shows up everywhere: the **surgical site infection** after a CS, **puerperal sepsis** after too many vaginal examinations with unwashed hands, a **neonatal sepsis** outbreak in the special care unit, and the **needle-stick** during a perineal repair at 2 a.m. On the paediatric rotation the same skills prevent cross-infection on the neonatal unit and the children's ward, where airborne (TB, measles) and contact (diarrhoea) precautions are daily decisions.

In Uganda, IPC is a matter of life and death, not paperwork. The country has repeatedly experienced **Ebola and Marburg outbreaks** in which health workers were among the first infected (from Dr Matthew Lukwiya in Gulu in 2000 to the Sudan-virus outbreaks of 2022 and 2025). HIV and hepatitis B are common, and antimicrobial resistance is rising sharply. Yet many facilities lack running water, alcohol rub, gloves or a functioning autoclave.

The one thing never to forget: *treat every patient's blood and body fluids as potentially infectious* (**standard precautions**), and **clean your hands at the 5 moments**. Hand hygiene is the single most effective IPC measure, and it costs almost nothing.

Definitions

Term	Meaning
Health-care-associated infection (HAI)	An infection that was not present or incubating on admission and appears ≥48 h after admission, or within 30 days of surgery (90 days if an implant is used)
Standard precautions	The minimum IPC practices for **all patients**, regardless of diagnosis
Transmission-based precautions	Extra measures, on top of standard precautions, for known or suspected infections: **contact, droplet, airborne**
Decontamination	The first step for used items: soaking in **0.5% chlorine** for 10 min to **inactivate HBV, HIV and HCV** and make items safer to handle
Cleaning	Physical removal of blood, tissue and dirt with **detergent, water and scrubbing**. Without it, no disinfectant or sterilant works reliably
High-level disinfection (HLD)	Kills all microorganisms **except some bacterial spores**
Sterilisation	Kills **all** microorganisms **including spores**
Antiseptic	A chemical used on **living tissue** (chlorhexidine, povidone-iodine, alcohol)
Disinfectant	A chemical used on **inanimate objects** (chlorine, glutaraldehyde, phenolics)
PEP	Post-exposure prophylaxis: a short course of ARVs (and HBV vaccine ± immunoglobulin) after an exposure to prevent infection
Outbreak	More cases than expected in a particular place and time, often with the same organism

Burden and Uganda context

- HAIs affect roughly **1 in 10 hospitalised patients in low- and middle-income countries**, about twice the rate in high-income countries (WHO). **Surgical site infection** is the commonest HAI in low-resource settings. After CS in Ugandan hospitals it runs at around **10–15%** or higher in several studies, dominated by **ESBL-producing *Klebsiella* and *E. coli* and MRSA**.
- **Puerperal sepsis** remains a leading direct cause of maternal death in Uganda. Many cases are preventable: unclean hands, frequent VEs, prolonged labour with unclean instruments, and unsafe abortion.

- **Neonatal sepsis outbreaks** in special care units are often traced to contaminated hands, shared equipment and multi-use vials.
- **Hand hygiene compliance** in Ugandan hospitals has been measured as **low at baseline**. Quality-improvement programmes (a WHO multimodal strategy with locally produced alcohol rub, training, audit and feedback) have shown large improvements in several hospitals.
- **Health worker exposure**: needle-stick injuries are very common among Ugandan interns, nurses and students, and **most are not reported**. Many health workers are **not fully vaccinated against hepatitis B**. Uganda has intermediate-to-high HBV prevalence (especially in the north and northeast).
- **Resources**: many HC IIIs and IVs have **intermittent water**, no functioning autoclave (so they rely on **boiling**), and irregular supplies of gloves and alcohol rub. So IPC in Uganda depends heavily on **0.5% chlorine** made from household bleach (**JIK**), boiling, and discipline.

Indications: standard precautions for everyone, extra precautions by route

Standard precautions (every patient, every time): hand hygiene, PPE by risk assessment, respiratory hygiene and cough etiquette, safe injections and sharps, environmental cleaning, linen and waste management, reprocessing of equipment, and patient placement.

Transmission-based precautions (added on top):

Type	Spread by	Examples	Key measures
Contact	Direct or indirect touch	MRSA/ESBL wound infection, diarrhoea, scabies, **Ebola/Marburg** (body fluids)	Gloves and gown for contact; dedicated equipment; isolation or cohorting
Droplet (>5 µm, travel about 1–2 m)	Coughing, sneezing, talking	Influenza, COVID-19, meningococcus, pertussis	**Medical mask** within 1–2 m; patient wears a mask; spacing
Airborne (<5 µm, remain suspended)	Inhalation over distance	**Pulmonary TB**, measles, varicella	**N95/FFP2 respirator** for staff; patient masked; well-ventilated or single room; open windows

Choosing PPE by risk assessment

Task	Gloves	Apron/gown	Mask	Eye protection
Taking BP, abdominal palpation (intact skin)	No	No	No	No
Venepuncture, cannulation	**Yes** (non-sterile)	No	No	No
Vaginal examination in labour	**Sterile** (or HLD)	Apron if splash risk	No	No
Conducting a delivery / perineal repair	**Sterile** (elbow-length if available)	**Plastic apron**	**Yes**	**Yes** (liquor and blood splash)
Caesarean section	Sterile, double	Sterile gown + apron	Yes	Yes
MVA / evacuation	Sterile/HLD	Apron	Yes	Yes
Handling placenta, cleaning spills, instruments	**Utility gloves**	Apron	Mask if splash	Yes if splash
Suspected **TB**	Standard	Standard	**N95** (staff); medical mask (patient)	As needed

Task	Gloves	Apron/gown	Mask	Eye protection
Suspected **Ebola/Marburg** (VHF)	**Double gloves**	**Coverall/gown + apron**	**N95 or medical mask per protocol**	**Face shield + goggles**, hood, boots. **Trained buddy** for doffing

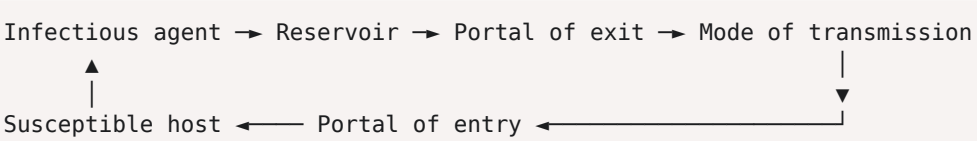
2. Contraindications and Precautions

> NOTE
> **Gloves do not replace hand hygiene.** Gloves have micro-holes, and hands get contaminated when gloves are removed. Clean your hands **before** putting on gloves (if a moment applies) and **always after** removing them. **Never wear the same gloves between two patients**, and never wash or alcohol-rub gloves to reuse them.

- **Alcohol rub is not enough** when hands are **visibly soiled** or contaminated with body fluids, after using the toilet, and with spore-forming organisms (e.g. *C. difficile*) or during diarrhoeal outbreaks: use **soap and water**.
- **Chlorine**: **never mix with acids or ammonia-containing cleaners** (releases toxic chlorine or chloramine gas). Do not soak instruments for **longer than 10 minutes** (corrosion). Do not use bleach on wounds or in the eye.
- **Glutaraldehyde** is toxic: use with ventilation and gloves; rinse items with boiled (HLD) or sterile (sterilisation) water.
- **Dry heat** only for glass and metal.
- **HLD is not sterilisation**: critical items (entering sterile tissue) must be sterilised wherever possible.
- **PEP is not indicated** for intact-skin exposure, for urine, faeces, saliva or vomit **without visible blood**, or **more than 72 hours** after the exposure. An **HIV-positive** worker needs ART, not PEP.
- **No routine antibiotics** for uncomplicated vaginal birth or episiotomy (WHO), and **do not prolong** surgical prophylaxis.
- **No routine environmental swabbing**: only when the epidemiology points to a source.
- **Do not aspirate or decant waste** or hand-sort bins; never recap needles by hand.

3. Relevant Anatomy and Normal Values

The chain of infection



Every IPC measure **breaks a link**:

Link	Obstetric example	How we break it
Agent	*Staph. aureus*, *E. coli*, GBS, HIV, HBV	Antimicrobials; sterilisation
Reservoir	Patient's skin and vagina, contaminated instruments, the environment	Cleaning, disinfection, sterilisation
Portal of exit	Blood, liquor, lochia, respiratory droplets	Cover wounds, respiratory hygiene, safe waste
Mode of transmission	**Hands (the commonest)**, instruments, sharps, droplets, air	**Hand hygiene**, PPE, safe sharps, isolation

Link	Obstetric example	How we break it
Portal of entry	Surgical wounds, cannula sites, the placental bed, perineal tears, needle-stick	Asepsis, sterile gloves for VE, careful technique
Susceptible host	Anaemic, exhausted, post-op, HIV-positive mothers; preterm babies	Nutrition, anaemia correction, vaccination (HBV, tetanus), prophylactic antibiotics when indicated

Spaulding classification (how clean must an item be?)

Category	Contact	Examples in O&G	Minimum processing
Critical	Enters **sterile tissue or the vascular system**	CS instruments, needles, suture needles, uterine sound, curettes	**Sterilisation**
Semi-critical	Touches **mucous membranes or non-intact skin**	**Vaginal specula**, MVA cannulae and aspirators, laryngoscope blades, delivery-set scissors (HLD acceptable where sterilisation is unavailable)	**HLD** at minimum (sterilise if possible)
Non-critical	Touches **intact skin** only	BP cuffs, stethoscopes, Pinard stethoscopes, beds, mackintoshes	**Cleaning** ± low-level disinfection

Why each step of instrument processing matters

1. **Decontamination (0.5% chlorine, 10 min)** makes the item **safe for the person who cleans it**. It inactivates HBV and HIV. **Not longer than 10 minutes**, because chlorine corrodes steel.
2. **Cleaning** removes up to about 99% of organisms and, crucially, the **organic material (blood, tissue) that shields microbes** and inactivates disinfectants. **"You cannot sterilise dirt."**
3. **HLD or sterilisation** kills what remains. Choose by the **Spaulding category** and by what is available.
4. **Storage** keeps it clean. A sterile item that is stored wet, in a torn wrap, or opened and left exposed is **no longer sterile**.

Why alcohol rub is preferred for routine hand hygiene

Alcohol rub (60–80% alcohol) **kills faster** than soap and water, takes **20–30 seconds**, needs **no sink or towel**, can be carried to the bedside, and is **kinder to skin** (it contains an emollient such as glycerol). Soap and water physically removes organic material and **spores** (alcohol does not kill *Clostridioides difficile* spores), so use it when **hands are visibly soiled**, after using the toilet, and during diarrhoeal outbreaks.

Key parameters ("normal values" for IPC)

Item	Value
Alcohol hand rub	**20–30 seconds**, about 3 mL (a palmful)
Soap and water handwash	**40–60 seconds**
Decontamination	**0.5% chlorine for 10 minutes**
HLD by boiling or steaming	**20 minutes** timed from the rolling boil
HLD by chemical soak	**0.1% chlorine** or **2% glutaraldehyde** for **20 min**
Autoclave	**121 °C, 106 kPa (15 lb/in²), 20 min unwrapped / 30 min wrapped**
Dry heat	**170 °C for 60 min** (or 160 °C for 120 min)
Chemical sterilisation	**2% glutaraldehyde for 8–10 hours**
Safety box	Close at **three-quarters full**

Item	Value
Droplet vs airborne particle size	Droplets **>5 µm** (travel 1–2 m); airborne **<5 µm**
Protective anti-HBs	**≥10 mIU/mL**
Needle-stick transmission risk	HBV **up to 30%** (HBeAg+), HCV **about 1.8%**, HIV **about 0.3%** (mucous membrane about 0.09%)
Sepsis warning signs (obstetric)	Temperature **≥38.0 °C or <36 °C**, pulse **>100/min**, RR **>20/min**, SBP **<90 mmHg**, confusion, urine **<30 mL/h**

4. Equipment and Preparation

Equipment

Item	Use	Notes
Alcohol-based hand rub (ABHR)	Routine hand hygiene	WHO formulation: ethanol 80% or isopropyl alcohol 75% + glycerol 1.45% + hydrogen peroxide 0.125%. Many Ugandan hospitals make their own
Liquid soap, running water, single-use towels	Handwashing when visibly soiled	**Bar soap in a wet dish and shared cloth towels recontaminate hands**
Examination gloves (non-sterile)	Contact with blood or body fluids, mucous membranes, non-intact skin	Single use. **Change between patients**, and do hand hygiene after removal
Sterile (surgical) gloves	VE in labour, delivery, procedures, surgery	HLD gloves are an acceptable alternative in low-resource settings
Utility (heavy-duty) gloves	Cleaning instruments, handling waste, cleaning spills	Reusable after washing
Plastic apron / fluid-resistant gown	Protect clothing and skin from splashes	Aprons are essential at every delivery
Medical (surgical) mask	Droplet precautions; splash protection; theatre	Covers **nose and mouth**
N95 / FFP2 respirator	Airborne precautions (TB, measles), aerosol-generating procedures	Needs a **fit check** each time
Goggles / face shield	Eye and mucous membrane protection from splashes (delivery, CS, cord cutting, placenta handling)	Often forgotten, yet delivery is a major splash risk
Gumboots / closed shoes	Protect feet from spills and dropped sharps	
Safety box (sharps container)	Immediate disposal of sharps	At the point of use; closed at three-quarters full
Colour-coded bins with liners	Waste segregation	Black, green, yellow, red, brown
Plastic buckets with 0.5% chlorine	Decontamination of instruments and gloves	Prepared daily
Autoclave / pressure steamer, drum or boiler	Sterilisation / HLD	Record every cycle; use **chemical indicator tape**

How to make 0.5% chlorine

Source	Concentration	How to dilute to 0.5%
JIK / household bleach	About **3.5%** available chlorine (check the label)	**1 part bleach + 6 parts clean water**. Formula: total parts water = (% concentrate ÷ % desired) − 1 = (3.5 ÷ 0.5) − 1 = 6
HTH / calcium hypochlorite powder	70%	**About 7 g (half a tablespoon) per litre of water**. Formula: g/L = (% desired ÷ % concentrate) × 1000 = (0.5 ÷ 70) × 1000 ≈ 7.1
Chlorine tablets (NaDCC)	Per manufacturer	Follow the package instructions

> NOTE
> Prepare chlorine solutions **daily** (or sooner if visibly dirty), because chlorine is unstable and weakens in light and heat. Store them in covered, opaque containers. **Never mix chlorine with acids or ammonia-containing cleaners**, because this releases toxic chlorine or chloramine gas. Use **0.1% chlorine** (JIK 1 part : 34 parts water) for **HLD by chemical soak** and for rinsing some surfaces.

Personal preparation

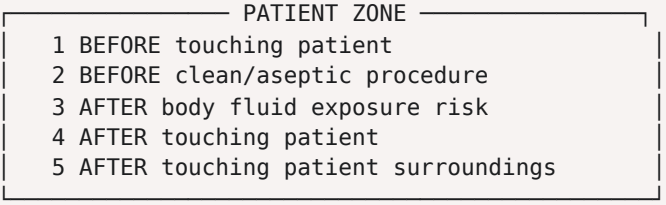
- **Bare below the elbows**, **no rings, watches or bracelets**, short nails, no artificial nails. Cover cuts with a waterproof dressing.
- **Hepatitis B vaccination** completed (3 doses at 0, 1 and 6 months), with the anti-HBs response checked where possible.
- Know your ward's **IPC focal person**, where the **PEP starter packs** are, and where the safety boxes and colour-coded bins are.

Consent and communication

- **Source testing after an exposure** needs the source patient's **informed consent** (rapid HIV, HBsAg, HCV) with pre-test counselling; keep the source identified (name and bed) so this can happen quickly.
- **Isolation or cohorting**: explain to the patient and family why precautions are needed, without stigma.
- **Placentas**: ask the family's wishes before disposal (see the waste section).

5. Procedure Step by Step

A. Hand hygiene: the WHO 5 Moments



Moment	When	Why	Labour-ward example
1. Before touching a patient	Before contact	Protects the **patient** from organisms on your hands	Before palpating the abdomen

Moment	When	Why	Labour-ward example
2. Before a clean/aseptic procedure	Immediately before, **even if you will wear gloves**	Protects the patient from organisms entering the body	Before a VE, inserting a catheter or cannula, perineal repair
3. After body-fluid exposure risk	Immediately after, **after glove removal**	Protects **you and the environment**	After the delivery, after emptying a urine bag, after handling the placenta
4. After touching a patient	When leaving the patient	Protects you and the environment	After the ANC examination
5. After touching patient surroundings	Even if you did not touch the patient	Protects you and the environment	After adjusting the drip or the bed linen

Alcohol hand rub technique (20–30 seconds):

1. Apply **a palmful** (about 3 mL) in a cupped hand, enough to cover all surfaces.
2. Rub **palm to palm**.
3. **Right palm over the left dorsum** with interlaced fingers, and vice versa.
4. **Palm to palm with fingers interlaced.**
5. **Backs of fingers** to the opposing palm with fingers interlocked.
6. **Rotational rubbing of the left thumb** clasped in the right palm, and vice versa.
7. **Rotational rubbing of the fingertips** of the right hand in the left palm, and vice versa.
8. Continue until the hands are **dry**. Then they are safe.

Handwashing with soap and water (40–60 seconds): wet hands, apply soap, perform the same 6 rubbing steps, rinse, **dry thoroughly with a single-use towel**, and **use the towel to turn off the tap**.

B. Donning and doffing PPE

Donning (put on) sequence:

1. **Hand hygiene.**
2. **Gown or apron**: fully cover the torso from neck to knees; tie at the back.
3. **Mask or respirator**: secure the ties or bands, fit the flexible band to the **nose bridge**, and fit snugly under the chin. For an N95, do a **seal check** (breathe out, and no air should leak at the edges).
4. **Goggles or face shield**: over the face and eyes, adjusted to fit.
5. **Gloves**: extend them **over the cuffs** of the gown.

Doffing (take off) sequence (the principle: the outside is contaminated, so touch only the "clean" inside and ties, and remove the **most contaminated items first**):

1. **Gloves**: grasp the outside of one glove at the wrist and peel it off inside out, holding it in the gloved hand. Slide the **ungloved fingers under the remaining glove at the wrist** and peel it off over the first. Discard in the **yellow bin**.
2. **Hand hygiene** (if the hands may be contaminated).
3. **Goggles or face shield**: remove from the back by the band or earpieces, without touching the front.
4. **Gown or apron**: unfasten the ties, pull away from the neck and shoulders touching the **inside only**, turn it **inside out**, fold or roll it into a bundle, and discard. (Alternatively remove the gown and gloves together, rolling them inside out.)
5. **Mask or respirator**: grasp the bottom ties or elastics, then the top ones, and remove **without touching the front**. For airborne precautions, remove the respirator **after leaving the room**.
6. **Hand hygiene** immediately after removing all PPE.

C. Processing instruments

Step 1: Decontamination (at the point of use, immediately after the procedure)

- Put on utility gloves. Place **all instruments, open**, in a **plastic bucket of 0.5% chlorine** so they are fully submerged. **Soak for 10 minutes**, then remove and rinse with water. Never leave instruments for hours, because chlorine corrodes metal.
- Wipe surfaces (couch, mackintosh, trolley) with 0.5% chlorine.
- Immerse reusable gloves (before removal) and dispose of single-use gloves in the yellow bin.

Step 2: Cleaning

1. Wear **utility gloves, apron and eye protection**.
2. Scrub every instrument **under the water surface** (to avoid splashing) with a soft brush, **detergent** and water. Open the hinges and clean the jaws, ratchets and teeth.
3. **Rinse** thoroughly with clean water.
4. **Dry** (air dry or with a clean towel). Wet instruments dilute chemical disinfectants and rust in storage.
5. Inspect for residual blood and damage.

Step 3a: High-level disinfection (HLD)

Method	How	Notes
Boiling	Fully submerge, lid on; **start timing when the water reaches a rolling boil**; boil **20 minutes**; don't add anything once timing has started	The most common HLD in Ugandan health centres
Steaming	**20 minutes** in a steamer	Good for gloves and MVA cannulae
Chemical	Soak in **0.1% chlorine for 20 min** or **2% glutaraldehyde for 20 min**; then **rinse with boiled water**	Glutaraldehyde is toxic (ventilation, gloves). Concentrations for chlorine HLD vary: general IP manuals use 0.1% (made with boiled water), while Ipas MVA processing guidance uses **0.5% for 20 min**. Follow your unit's protocol

Use HLD items immediately, or store them in a covered HLD container for up to about 1 week.

Step 3b: Sterilisation

Method	Parameters	Notes
Steam autoclave	**121 °C, 106 kPa (15 lb/in²): 20 minutes unwrapped, 30 minutes wrapped**	The gold standard. Use **chemical indicator tape** on each pack, record the cycle, and use periodic biological indicators where available
Dry heat (oven)	**170 °C for 60 minutes** (or 160 °C for 120 minutes)	Only for glass and metal. Timing starts when the temperature is reached
Chemical sterilisation	**2% glutaraldehyde for 8–10 hours**, then rinse with sterile water	When heat is not possible

Step 4: Storage: wrapped sterile packs go on clean, dry, covered shelves off the floor. The pack is sterile until it is **opened, torn, wet or dropped** (event-related sterility). Label every pack with the date and "sterilised by".

D. Health-care waste segregation (Uganda MoH colour code)

Container	Waste	Examples in Ob/Gyn	Final disposal
Black bin / liner	**Non-infectious general waste**	Paper, packaging, food leftovers	Municipal collection / burning
Green bin / liner	**Non-infectious plastics** (recyclable) Verify *green category and its use differ between MoH documents and facilities*	Clean drug packaging, bottles	Recycling / municipal
Yellow bin / liner	**Infectious waste**	Blood-soaked gauze, swabs and pads, used gloves, IV giving sets, catheters	Incineration
Red bin / liner	**Highly infectious waste**	Laboratory cultures and specimens, waste from isolation of highly infectious diseases (Ebola/VHF); anatomical waste per facility policy	Incineration (after autoclaving where required)
Yellow safety box	**Sharps**	Needles, blades, suture needles, broken ampoules, cannula stylets	Close at **three-quarters full**, seal, incinerate
Brown bin / liner	**Pharmaceutical and chemical waste**	Expired drugs, chemicals	Return to NMS or special disposal

UGANDA CONTEXT

Placentas: in Uganda many families wish to take the placenta home for culturally important burial. Respect this. Hand it over in a **sealed, leak-proof bag**, and explain how to handle it (gloves or bag, and burying it deep). Otherwise, dispose of placentas in the **placenta pit** or incinerator per facility policy. Do **not** put placentas in the black general bin.

Segregation rules: segregate **at the point of generation** (at the bedside, not later), fill bins to **no more than three-quarters**, never decant or hand-sort waste, and label bins with the ward and date.

E. Safe sharps practice

1. **Minimise injections**: use oral drugs where possible.
2. **Never recap needles** by hand. If recapping is unavoidable, use the **one-handed scoop technique**.
3. **Don't bend, break or remove needles** from syringes by hand.
4. **Dispose of the sharp immediately, at the point of use, by the user**, into a safety box within arm's reach.
5. Use a **neutral zone (kidney dish)** for passing sharps in theatre and at delivery. Announce "sharp".
6. Suture with **instruments, not fingers** (use forceps to hold tissue and to handle the needle). During perineal repair, don't feel for the needle tip with a finger.
7. **Double gloving** at CS and major surgery.
8. **Get vaccinated against hepatitis B** (3 doses at 0, 1 and 6 months), and check the anti-HBs response where possible.

F. Environmental cleaning and spills

- Clean **from clean to dirty and top to bottom**. Use detergent and water for routine cleaning, then disinfect high-touch and blood-contaminated surfaces.

- **Blood or body-fluid spill**: put on utility gloves (and apron and eye protection if large). **Cover the spill with 0.5% chlorine** (or absorbent material soaked in chlorine), leave for about **10 minutes**, wipe up with disposable cloths into the yellow bin, clean with detergent and water, then dry.
- After **each delivery**: clean and disinfect the delivery bed and mackintosh, change the linen, and clean the floor under the bed.
- **Linen**: handle soiled linen with gloves, **don't shake** it, bag it at the bedside, and transport it in leak-proof bags. Heavily soiled linen is soaked in 0.5% chlorine, then washed.

G. IPC in the labour ward (putting it together)

Practice	Why
Hand hygiene at the 5 moments, especially **before every VE**	Hands carry *E. coli*, *Staph* and GBS into the genital tract
Sterile (or HLD) gloves for VE; limit VEs to **every 4 hours** in the active first stage unless indicated	Each VE after ROM increases the risk of chorioamnionitis
Clean perineum with **clean water** before VE and delivery	Mechanical cleansing
Sterile delivery set for each woman (scissors, cord clamps, artery forceps)	Tetanus and sepsis prevention (maternal and neonatal)
Full PPE for delivery: apron, gloves, mask and **eye protection**, boots	Splashes of liquor and blood are common; HIV/HBV
Antibiotic prophylaxis only when indicated: CS (before incision), manual removal of the placenta, 3rd/4th-degree tears	WHO does **not** recommend routine antibiotics for uncomplicated vaginal birth or episiotomy (resistance)
Vaginal cleansing with povidone-iodine before CS in labour or with ROM	Reduces endometritis
Clean cord care: chlorhexidine 7.1% digluconate (4% chlorhexidine) gel to the cord stump (Uganda policy); nothing else on the cord	Prevents omphalitis and neonatal sepsis. No cow dung, herbs or ash
Tetracycline 1% eye ointment for every newborn	Prevents ophthalmia neonatorum
Skin-to-skin, early exclusive breastfeeding	Colonises the baby with maternal flora and provides immune protection
Clean the bed, mackintosh and floor after every birth	Environmental reservoir
One woman per bed	Overcrowding (two mothers per bed) spreads infection

H. Surgical site infection prevention bundle (WHO)

Before surgery: the patient bathes with soap; **no shaving** (clip if necessary); treat remote infections; correct anaemia and control blood sugar; give **antibiotic prophylaxis before incision** (for CS, within 60 min before the skin incision).

In theatre: surgical hand antisepsis; **chlorhexidine-alcohol skin prep**; sterile drapes and gowns; minimal traffic; **gentle tissue handling**; normothermia; **FiO_2 80%** under general anaesthesia with intubation (WHO recommendation); change gloves after vaginal manipulation; consider **triclosan-coated sutures** where available.

After surgery: **do not prolong prophylactic antibiotics**; use a sterile dressing for 24–48 h; hand hygiene before touching the wound; watch for SSI (day 3–7) and educate the woman about it.

I. Examining a patient with a suspected HAI (Hutchison order)

1. **Introduction and consent**; explain; chaperone for breast and perineal examination.
2. **Hand hygiene**; gloves and apron if the wound is draining (contact precautions).
3. **Position and exposure**: semi-recumbent; expose each area in turn, preserving dignity.
4. **General inspection and vital signs**: well or unwell, toxic, pallor, dehydration; temperature, pulse, BP, RR, SpO_2, conscious level, urine output; compare with the sepsis warning signs (section 3) and the obstetric early-warning chart.
5. **Hands and arms**: cannula sites (phlebitis).
6. **Face and neck**: mucosa, hydration, lymph nodes.
7. **Chest**: crackles, reduced air entry (hospital-acquired pneumonia); **breasts** (mastitis).
8. **Abdomen**: inspect and palpate the **wound**; palpate the **uterus** (tenderness, involution).
9. **Perineum and lochia**: repair site; offensive lochia.
10. **Catheter and urine**: cloudy urine, suprapubic pain.
11. **Calves**: swelling, tenderness (DVT as a non-infective cause of fever).
12. **Complete**: thank the patient, remove PPE in order, **hand hygiene**, summarise and plan.

6. Findings and Interpretation

The patient with a suspected HAI: what each finding means

Site	What to look for	Suggests
Cannula sites	Red, tender, cord-like vein	Phlebitis, line infection
Chest	Crackles, reduced air entry	Hospital-acquired pneumonia
Breasts	Red, tender, engorged	Mastitis
Wound	Redness, induration, pus, gaping, crepitus	SSI (superficial, deep), necrotising infection
Uterus and lochia	Tender uterus, **offensive lochia**	Endometritis (organ/space SSI)
Perineum	Swollen, discharging repair	Perineal wound infection
Catheter and urine	Cloudy urine, suprapubic pain	Catheter-associated UTI
Calves	Swelling, tenderness	DVT (a non-infective cause of fever)

The exposure site

Look at the **wound** (depth, bleeding, contamination) or the splashed eye or mouth: puncture, laceration, bleeding, skin integrity. Check the exposed person's general health (for the baseline).

"Examine the ward" (the environmental walk-through during an outbreak)

Check	Acceptable	Breach
Alcohol rub at the point of care and **working sinks**	At each bed or on the trolley; soap and single-use towels	Missing rub; bar soap; shared towels
Chlorine buckets	Made **today**, correct strength, covered	Made weekly; uncovered; wrong dilution
Autoclave	Logbook complete, **indicator tape** on every pack, cycle times correct	No tape; gaps in the log
Sterile store	Packs dry, intact, in date, off the floor	Wet, torn or undated packs

Check	Acceptable	Breach
Safety boxes	Within reach, **not more than three-quarters full**	Overfilled; sharps in general bins
Waste bins	Colour-coded and used correctly	Gloves in the black bin
Multi-dose vials	Labelled, dated, accessed with a new needle each time	Undated; shared needles
Bed occupancy, shared equipment	One woman per bed; equipment cleaned between patients	Two women per bed; shared suction or bag-mask
Staff practice	Hand hygiene at the 5 moments; gloves changed between patients	Gloves worn patient to patient

Red-flag findings on equipment

- **Autoclave indicator tape not changed colour, or a wet or torn pack**: **not sterile**. Do not use it.
- **Overfilled safety box, or sharps in a general bin**: a hazard to cleaners and waste handlers.

Bedside tests

Test	Use
Rapid HIV test (exposed worker, baseline, with counselling)	A positive result means ART, not PEP. Do it **the same day**
Rapid HIV test on the source (with consent)	A negative source (and no recent high-risk exposure) means PEP can be stopped
Rapid HBsAg test (source and worker)	Guides the hepatitis B plan
Urine pregnancy test (exposed woman)	Counselling; TLD is safe in pregnancy
Temperature, pulse, BP trend (patient)	Sepsis recognition and response to treatment
HemoCue Hb, RBS, urine dipstick, malaria RDT (patient)	Anaemia and high glucose delay healing; dipstick for UTI; **malaria is a common cause of fever in Uganda**
Hand hygiene observation audit (WHO observation form, 20–30 minutes on the ward)	Measures compliance at the 5 moments; gives immediate feedback
Check the chemical indicator on packs in use	An unchanged indicator means a sterilisation failure

Laboratory and imaging

Test	When	Expected / abnormal finding	How it changes management
HIV test of the exposed worker (rapid test at all levels)	**Baseline**, then about **6 weeks** and **3 months** (some protocols add 6 months)	Negative expected. A positive baseline = infection **before** the exposure	Positive baseline: stop PEP, start ART. Seroconversion: ART and incident review
Source HIV test and viral load (rapid test at all levels; VL from the ART clinic records or CPHL samples)	At the time of exposure, with consent	Negative, or positive with suppressed VL, lowers the risk	Negative source: PEP can stop. Positive or unknown: complete 28 days
HBsAg (source and worker; rapid test at HC III and above)	Baseline	Source HBsAg-positive	Decides HBIG and vaccine for a non-immune worker
Anti-HBs level (regional/national referral or private labs)	Vaccinated worker with an unknown response	**≥10 mIU/mL = protected**	<10: HBIG + booster (section 8)

Test	When	Expected / abnormal finding	How it changes management
HCV antibody; HCV RNA (referral/private labs)	Baseline and follow-up if the source is HCV-positive or unknown	Seroconversion	Refer for direct-acting antiviral treatment
Creatinine (hospital labs)	Baseline for TDF if available	Normal	**Not needed before the first PEP dose**; adjust if renal impairment
Pus swab for culture and sensitivity (RRH and national referral microbiology; rarely HC IV)	Every infected wound, **before antibiotics**	**ESBL *Klebsiella/ E. coli*, MRSA**, *Pseudomonas*	Targeted antibiotics; the same organism in several patients supports a common source
Blood cultures (RRH/national)	Fever with sepsis signs	Bacteraemia	Targeted antibiotics; cluster detection
Urine culture (RRH/national)	Suspected CAUTI	Significant growth	Remove or change the catheter; antibiotics
FBC, CRP (hospital labs)	Suspected infection	Raised WBC (remember it is higher in labour and after steroids)	Supports infection; follow the trend
Ultrasound (hospitals)	Fever not settling after 48–72 h	Pelvic or wound collection, retained products	Drainage or evacuation
Biological indicator (spore test) of the autoclave (where available; national/regional sterile services)	During an SSI outbreak, and routinely	Growth = sterilisation failure	Stop the autoclave, recall packs, repair
Environmental swabs (national/regional microbiology)	**Only when the epidemiology points to a source** (e.g. a shared suction machine)	The outbreak organism on equipment or fluids	Remove or reprocess the source. Routine environmental swabbing is **not** recommended
Organism typing (national reference laboratories; rarely available)	Complex outbreaks	Same strain in several patients	Confirms an outbreak, but **you do not need it to start control measures**

Decision grid for HIV PEP

Exposure	Source HIV-positive	Source unknown	Source HIV-negative
Percutaneous, mucous membrane or non-intact skin, with blood or infectious fluid	**PEP**	**PEP** (review when the source result is back)	No PEP (consider the window period if the source is at high risk)
Intact skin, or a non-infectious fluid (urine, vomit without blood)	No PEP	No PEP	No PEP

7. Clinical Acumen

IPC acumen is about **risk assessment and noticing**: seeing the missed moment, predicting the exposure, and knowing what to do when something goes wrong. There are two situations you must be able to work up step by step: **(A) an occupational exposure** (a needle-stick, cut or blood splash to you or a colleague), and **(B) a suspected health-care-associated infection or outbreak** (for example a woman with a post-CS wound infection, or several women or babies infected on the same ward).

Focused history: A. after an exposure (each answer changes the PEP decision)

Question	Why it matters
When did it happen?	PEP ideally within **2 h**, never after **72 h**
Type: percutaneous, mucous membrane, non-intact skin, or intact skin	**Intact skin exposure needs no PEP**
Device and depth: **hollow-bore needle** (cannula, venepuncture) vs solid suture needle; deep injury; **visible blood** on the device	**Hollow-bore, deep and visibly bloody** injuries carry the highest risk
Body fluid: blood, bloody fluid, **liquor, vaginal secretions** (infectious) vs urine, faeces, vomit without visible blood (low risk)	Decides whether the exposure is significant
Glove worn? Double gloves?	A glove wipes off some of the blood and lowers the dose
Source: HIV status, **on ART? last viral load?** HBsAg, HCV	A suppressed source gives a very low risk, but an unknown or unsuppressed source means PEP; **never delay PEP waiting for source results**
Your own HIV status (last test), **hepatitis B vaccination** (number of doses) and **anti-HBs** if known	An HIV-positive worker needs ART, not PEP; an unvaccinated worker needs HBV vaccine ± HBIG
Pregnancy or breastfeeding	TLD is safe in pregnancy; it affects counselling
Other medicines, kidney disease	TDF (renal); DTG interacts with polyvalent cations: take DTG 2 h before or 6 h after iron, calcium or antacids

Focused history: B. suspected HAI or outbreak (ask of each case)

Question	Why it matters
What procedure and when? Date of CS or delivery, theatre, operator, **day of onset** of infection	SSIs appear mostly on **day 4–7**; a common theatre, day or team points to a common source
Antibiotic prophylaxis: which drug, and when?	Given after cord clamping or not at all means less protection
Labour details: duration, ROM duration, **number of VEs**, prolonged obstructed labour	Host risk vs system failure
Skin preparation, shaving, wound dressing, who changed it	Common breaches
For a neonatal cluster: shared equipment (suction, bag-mask, incubators), **multi-dose vials**, how IV fluids and feeds are prepared, hand hygiene before handling babies	Typical sources of Gram-negative neonatal outbreaks
Recent changes: new staff, broken autoclave, water or chlorine shortage, new supplier	Outbreaks often follow a system change
Is she the only one?	Other similar infections recently suggest an **outbreak**

Red flags

RED FLAG

- **Needle-stick or splash from any patient**: report and assess **immediately**. The PEP window is closing.
- **A cluster** of SSIs, puerperal sepsis or neonatal sepsis in the same ward in a short time: an **outbreak**. Notify the IPC focal person urgently.
- **Fever with bleeding, from an area with an ongoing Ebola/Marburg outbreak** or contact with a sick or dead person: **isolate, full VHF PPE, notify the district immediately**. Do not examine vaginally or deliver without full PPE.
- **Cough for 2 weeks or more with weight loss in a ward patient**: suspect TB. Mask the patient, give staff N95s, and investigate (GeneXpert).
- **Autoclave indicator tape not changed colour, or a wet or torn pack**: **not sterile**. Do not use it.
- **Overfilled safety box, or sharps in a general bin**: a hazard to cleaners and waste handlers.

Thinking like a clinician

1. **Assume every patient is infectious.** You cannot tell HIV or HBV status by looking, and many women are in the window period at delivery. That is the logic of **standard precautions**.
2. **Match processing to the Spaulding category**: a speculum (mucous membrane) needs **HLD**; a curette entering the uterus (sterile cavity) needs **sterilisation**. When an autoclave is unavailable, HLD by boiling is the accepted fallback for semi-critical items.
3. **"If X then think Y"**:
   - A foul-smelling SSI 5 days after CS with a **cluster** of similar cases means an IPC breach (theatre, antibiotic timing, the autoclave). Audit, don't just treat.
   - A health worker who is **unvaccinated for HBV** after a needle-stick: HBV is **100 times more infectious** than HIV. Give the HBV vaccine (± HBIG) urgently.
   - A **suppressed source viral load**: the HIV transmission risk is negligible, but most Ugandan protocols still give PEP if there is any doubt or the VL result is old.
4. **Common student mistakes**:
   - Wearing gloves from patient to patient ("I'm wearing gloves, so it's fine").
   - Skipping moment 2 because "I'll wear sterile gloves anyway".
   - Wearing a mask below the nose. Not using eye protection at deliveries.
   - Recapping needles.
   - Delaying reporting a needle-stick until the end of the shift, or not reporting at all ("it was only a suture needle").
   - Leaving instruments in chlorine all day, or skipping cleaning and "sterilising" dirty instruments.
   - Throwing gloves into the black bin.

Putting it together

A. Model summary of an exposure: "A 23-year-old student sustained a **deep percutaneous injury** with a **hollow-bore cannula needle, visibly bloody**, at 14:10, from a woman of **unknown HIV status**. First aid done. Worker HIV-negative at baseline, **3 doses of HBV vaccine, anti-HBs unknown**. Assessment: **significant exposure, source unknown**. Plan: **TLD 1 tablet daily for 28 days, first dose now (within 2 hours)**, source HIV and HBsAg tests with consent, worker anti-HBs, follow-up at 2 weeks, HIV tests at 6 weeks and 3 months, incident report."

Worked example 1: the source result comes back. The student above started TLD at 14:40. At 16:00 the source's rapid HIV test is **reactive** and she tells you she stopped ART a year ago; the HBsAg is **negative**. The student's anti-HBs is later **25 mIU/mL**. *Interpretation:* a high-risk HIV exposure (hollow needle, visible blood, **unsuppressed source**); **protected against hepatitis B**. *Plan:* **complete 28 days of TLD**, adherence support, review at 2 weeks, HIV tests at **6 weeks and 3 months**, condoms and no blood donation until the 3-month test is negative. No HBIG needed. Also link the **source woman back to HIV care** (restart ART, check viral load) and make an eMTCT plan for her baby.

B. Model summary of a suspected outbreak: "In the last 3 weeks, **5 women** developed **SSIs after CS** on our ward, compared with about 1 a month usually. **4 of 5 pus cultures grew ESBL *Klebsiella***. Onsets were on days 4–6. All were operated in theatre 2."

Worked example 2: interpreting the outbreak investigation.

- **Case definition**: "Any woman with a superficial, deep or organ/space SSI within 30 days of a CS done in our hospital from 1 to 21 March."
- **Line list**: name, age, date of surgery, theatre, surgeon, antibiotic and its timing, labour duration, onset day, culture result, outcome.
- **Findings of the audit**: antibiotic given **after cord clamping** in 4 of 5 cases; the theatre 2 autoclave has **no indicator tape in stock** for 2 weeks; chlorine buckets are made **weekly**; hand rub is missing from the post-op ward.
- *Interpretation:* more cases than expected, clustered in place and time, with the **same organism**: a likely **outbreak** with several system failures (prophylaxis timing, unverified sterilisation, weak decontamination and hand hygiene).
- *Control measures:* treat each woman (drain, culture-guided antibiotics); **contact precautions and cohorting**; **antibiotic before incision** at every CS (make it part of the Time out); stop using theatre 2's autoclave until verified, with indicator tape and a spore test; **fresh chlorine daily**; alcohol rub at every bed; **feed back** the results to staff; continue **SSI surveillance** (post-discharge follow-up to 30 days) to confirm the outbreak is over.

8. Complications and Troubleshooting

First 5 minutes after an exposure (needle-stick, cut or splash)

1. **Make the patient safe.** Say "sharp injury", hand over or finish the critical step (e.g. clamp a bleeder), and put the sharp in the **safety box**.
2. **First aid now**: let a percutaneous wound **bleed freely**, **do not squeeze or suck**, and **wash with soap and running water**. Flush eyes or mouth with **plenty of clean water or saline for 10–15 minutes**. No bleach or alcohol on the wound.
3. **Note the time.** PEP works best **within 2 hours** and must start **within 72 hours**.
4. **Report immediately** to the in-charge or IPC focal person. Do not wait until the end of the shift.
5. **Keep the source patient identified** (name and bed) so that consent and testing can be done quickly.

Needle-stick / sharps injury or splash: full algorithm

1. **First aid immediately**:
    - **Percutaneous (needle-stick, cut)**: let the wound **bleed freely**, **do not squeeze or suck**, and **wash with soap and running water** for several minutes. Don't use bleach, alcohol or other caustics on the wound.
    - **Mucous membrane (eye, mouth)**: **flush with plenty of clean water or saline** for 10–15 minutes. Remove contact lenses. Don't use soap in the eye.
    - **Intact skin**: wash with soap and water. No PEP needed.

2. **Report immediately** to the ward in-charge or IPC focal person, and **document** the incident.
3. **Assess the exposure risk**: the type (percutaneous or mucous membrane is significant; intact skin is not) and the material (blood, bloody fluid, liquor, vaginal secretions are infectious; urine, faeces or vomit without visible blood are low risk).
4. **Assess the source** (with informed consent): **rapid HIV test**, **HBsAg**, HCV antibody. Check records for known status and **viral load**. **Do not delay PEP while waiting for source results.**
5. **Assess the exposed worker**: **baseline HIV test** (with counselling), HBsAg and anti-HBs where available, HBV vaccination history, **pregnancy test** if relevant, creatinine if available (not required before the first dose).
6. **Start PEP as soon as possible**, ideally within **2 hours**, **no later than 72 hours**, if the exposure is significant and the source is HIV-positive or of unknown status.
7. **Hepatitis B**: see the table below.
8. **Counsel**: adherence (28 days), side effects, **condom use and no blood donation until the 3-month test is negative**, and psychological support.
9. **Follow up**: review at 2 weeks (adherence, side effects), and **repeat HIV testing at about 6 weeks and 3 months** after the exposure (some protocols add 6 months, especially with HCV co-exposure). Check HCV and HBV markers per exposure.
10. **Prevent recurrence**: review how it happened (a root-cause analysis without blame).

HIV PEP drug table (Uganda / WHO)

Drug	Dose	Route	Frequency	Duration	Notes
TDF/3TC/DTG (TLD): tenofovir DF 300 mg + lamivudine 300 mg + dolutegravir 50 mg	**1 tablet**	Oral	**Once daily**	**28 days**	**Preferred regimen for adults and adolescents** (WHO 2024; Uganda consolidated guidelines). Safe in pregnancy. Give the **full 28-day supply** at the first visit. Take DTG 2 h before or 6 h after iron, calcium or antacids
TDF/3TC + **ATV/r** (atazanavir 300 mg / ritonavir 100 mg)	1 tablet TDF/3TC + ATV/r	Oral	Once daily	28 days	Alternative if DTG is unavailable or not tolerated
TDF/3TC + LPV/r	LPV/r 400/100 mg	Oral	Twice daily	28 days	Alternative (more side effects)

DRUG DOSING

PEP is an emergency. Don't wait for the source's result, or for a doctor to "see you tomorrow". Every Ugandan hospital and ART clinic, and many casualty departments, keep **PEP starter packs**. The 72-hour limit is absolute. Beyond it, PEP is not recommended, but still test and follow up.

UGANDA CONTEXT

The Uganda **National IPC Guidelines** (MoH, 2013, with later editions) and the national health-care waste management guidelines set the standards you'll be examined on: **0.5% chlorine** for decontamination, the **autoclave cycle**, and the **colour-coded bins**. The national **PEP policy** makes PEP free at all public ART sites. Every hospital should have an **IPC committee and focal person**. Ask on the ward who yours is, because that's who you report a needle-stick to. Uganda's PEP regimen (TLD) matches WHO 2024; check that the HIV follow-up testing schedule in your unit matches the current consolidated guidelines Verify *Uganda follow-up test timings may differ from WHO 2024*.

Hepatitis B after exposure

Exposed worker	Source HBsAg positive (or unknown, high risk)	Source HBsAg negative
Unvaccinated / incomplete	**HBV vaccine dose 1 now** (complete 0, 1, 6 months) **+ HBIG** (hepatitis B immunoglobulin) as soon as possible, ideally within 24 h (up to 7 days) where available	Start or complete vaccination
Vaccinated, known responder (anti-HBs ≥10 mIU/mL)	No treatment	No treatment
Vaccinated, non-responder	HBIG + revaccination (or HBIG ×2)	Consider revaccination
Vaccinated, response unknown	Test anti-HBs. If <10, give HBIG + a vaccine booster	No treatment

Hepatitis C: there is no vaccine or PEP. Test at baseline and follow up (HCV RNA at about 3–6 weeks or antibody at 3–6 months). Refer for **direct-acting antiviral treatment** if infection occurs.

A woman with a suspected ward infection (first 5 minutes)

1. **Assess her first, not the paperwork**: airway, breathing, pulse, BP, RR, temperature, SpO_2, conscious level, urine output. **Sepsis warning signs** include temperature **≥38.0 °C or <36 °C**, pulse **>100/min**, RR **>20/min**, SBP **<90 mmHg**, confusion, or urine **<30 mL/h**. Use your unit's obstetric early-warning chart.
2. If she is septic: **cultures (blood, pus, urine) if they do not delay treatment**, then **IV antibiotics within the first hour** and IV fluids. Call a senior.
3. **Apply the right precautions straight away**: **contact precautions** (gloves and apron, dedicated equipment) for a draining wound or a suspected MDR organism; isolate or cohort if possible.
4. **Ask "is she the only one?"** If there have been other similar infections on the ward recently, it may be an **outbreak**: tell the ward in-charge and **IPC focal person today**.
5. **Red flag for VHF**: fever with unexplained bleeding plus travel to or contact with an outbreak area (Ebola/Marburg). **Isolate, full VHF PPE, notify the district at once.**

Managing a suspected HAI outbreak (e.g. cluster of post-CS SSIs)

1. **Recognise** it: more cases than expected in a place and time.
2. **Notify** the IPC focal person and committee and the ward leadership.
3. **Treat** patients (culture-guided antibiotics; drainage).
4. **Investigate**: case definition, line list, cultures (pus, blood), and **audit practices**: antibiotic timing, skin prep, theatre cleaning, the autoclave log and indicators, reused items, multi-dose vials, hand-rub availability.
5. **Control**: fix the breach (e.g. repair the autoclave, retrain staff, single-use vials), and cohort or isolate if MDR organisms are present.
6. **Monitor and feed back**: SSI surveillance and hand hygiene compliance audits.

Suspected Ebola/Marburg (VHF) in a pregnant woman (Uganda-specific)

1. **Isolate** her immediately in a designated area. **Limit staff** to trained personnel with **full VHF PPE**.
2. **Notify** the district surveillance focal person or MoH hotline immediately.
3. **Minimise invasive procedures**. Any delivery must be conducted in full PPE with safe disposal of all waste (**red**, highly infectious).
4. Monitor all exposed staff for 21 days.

(VHF in pregnancy carries very high maternal and near-total fetal mortality, and delivery is a major exposure event for health workers.)

Troubleshooting common IPC failures

Problem	Why it matters	Fix
No running water	Handwashing impossible	Alcohol rub; a tippy-tap or bucket with a tap; report
No alcohol rub	Moments missed	Locally produced WHO formulation; soap and water
Autoclave broken	Critical items cannot be sterilised	HLD by boiling for semi-critical items; send critical items to a working sterile service; do not use unverified packs
Chlorine made weekly or uncovered	Weak, ineffective solution	Make **daily**; covered, opaque containers
Instruments left in chlorine all day	Corrosion; damaged instruments	**10 minutes** only, then clean
Safety box full	Needle-stick risk to staff and cleaners	Close at three-quarters; replace; never compress
Gloves out of stock	Exposure risk	Prioritise sterile gloves for VE and delivery; report through the IPC committee

9. Documentation and Presenting Findings

What to document

- **Exposure incident report**: date and **time**, what happened (device, depth, visible blood, glove worn), body fluid, source (identified, consent obtained, test results), first aid given, worker baseline tests, **time PEP started**, HBV plan, follow-up dates, and a blame-free root-cause note.
- **Instrument processing**: every autoclave cycle in the **logbook**, indicator tape on every pack, each pack labelled with the **date and "sterilised by"**.
- **Outbreak**: the case definition, the line list, the audit findings, the control measures and the feedback given.
- **Waste**: bins labelled with the **ward and date**.

Model presentation (needle-stick incident, reporting to your supervisor)

> "Doctor, I need to report an occupational exposure. At 02:15, while repairing a second-degree tear on a 24-year-old P2 whose HIV status is positive (on TLD, last viral load 6 months ago undetectable), I sustained a **percutaneous injury with a solid suture needle through my glove**, with visible bleeding. I immediately let it bleed and washed it with soap and running water. My last HIV test was negative 3 months ago, and I had 3 doses of hepatitis B vaccine in first year but don't know my antibody level. I'm asking for **PEP (TLD for 28 days) now**, a baseline HIV test, anti-HBs, and an incident report."

10. OSCE Checklist

Station 1: "Demonstrate hand hygiene with alcohol rub"

Step	Done well when	Critical?
1. Preparation	Bare below the elbows; no jewellery; short nails	Yes
2. Indication	States which of the 5 moments applies	No

Step	Done well when	Critical?
3. Amount	Applies a palmful of rub to a cupped hand	No
4. Palm to palm	Rubs palm to palm	Yes
5. Dorsum	Right palm over left dorsum with interlaced fingers, and vice versa	Yes
6. Interlaced	Palm to palm, fingers interlaced	Yes
7. Backs of fingers	Backs of fingers to opposing palms, fingers interlocked	Yes
8. Thumbs	Rotational rubbing of each thumb	Yes
9. Fingertips	Rotational rubbing of fingertips in the opposite palm	Yes
10. Dry	Continues until dry (20–30 s); does not wipe or wave dry	Yes

Station 2: "Don and doff PPE for a delivery / isolation patient"

Step	Done well when	Critical?
1. Hand hygiene	Done first	Yes
2. Gown/apron	On and tied	No
3. Mask	On, nose bridge moulded, under the chin (seal check for N95)	Yes
4. Eye protection	Goggles/face shield on	No
5. Gloves	On, over the gown cuffs	No
6. Gloves off first	Glove-to-glove, skin-to-skin technique	Yes
7. Hand hygiene	After glove removal	Yes
8. Eye protection off	Removed from behind	No
9. Gown off	Touching the inside only, rolled inside out	Yes
10. Mask off	By the ties or elastics, without touching the front	Yes
11. Hand hygiene	At the end	Yes
12. Disposal	PPE into the correct (yellow) bin	No

Station 3: "Process this used delivery set" / "Make 0.5% chlorine"

Step	Done well when	Critical?
Decontaminate	Utility gloves; instruments open and submerged in 0.5% chlorine for 10 min	Yes
Clean	Under water, brush, detergent, hinges opened; rinse; dry	Yes
HLD or sterilise	Boil 20 min from the rolling boil, or autoclave 121 °C at 106 kPa for 20/30 min	Yes
Store	Dry, covered, labelled, off the floor	No
Chlorine	States the JIK dilution (1:6 for 3.5% JIK) and the formula	Yes

Station 4: "Your colleague has just had a needle-stick"

Step	Done well when	Critical?
First aid	Bleed freely, no squeezing, soap and running water	Yes
Report	Immediately, to the in-charge or IPC focal person	Yes
Assess	Exposure type and fluid; source status and viral load	Yes

Step	Done well when	Critical?
Baseline tests	Worker HIV, HBsAg, anti-HBs, pregnancy test; source tests with consent	No
PEP	**TLD for 28 days**, within 72 h (ideally 2 h), not delayed for the source result	Yes
Hepatitis B	Plan by vaccination status and anti-HBs	Yes
Counselling and follow-up	Adherence, condoms, no blood donation; tests at 6 weeks and 3 months	No

ll. Key Points

High-yield points

- **Standard precautions for every patient**; hand hygiene at the **5 moments** is the single most effective measure.
- **Gloves never replace hand hygiene.**
- **D-C-H/S-S**: decontaminate (0.5% chlorine, 10 min), clean, HLD or sterilise, store. You cannot sterilise dirt.
- **Spaulding**: critical → sterilise; semi-critical → HLD; non-critical → clean.
- **Bins**: black general, yellow infectious, red highly infectious, brown pharmaceutical; sharps in a safety box closed at three-quarters.
- **Needle-stick**: bleed, wash, report, PEP (TLD × 28 days) ideally within 2 h, never after 72 h; never wait for the source result.
- **Hepatitis B** is the biggest occupational threat and is vaccine-preventable.
- **Outbreak**: more cases than expected; notify, define, line-list, culture, audit, control, feed back.
- SSI isolates in Uganda are often **ESBL Klebsiella/E. coli and MRSA**; culture before antibiotics.

Numbers & doses

Item	Value
Alcohol hand rub	**20–30 seconds**
Soap and water handwash	**40–60 seconds** (when hands are visibly soiled, after the toilet, spores/diarrhoea)
Decontamination	**0.5% chlorine for 10 minutes** (no longer: corrosion)
0.5% chlorine from 3.5% JIK	**1 part JIK + 6 parts water**
0.5% chlorine from HTH (70%)	About **7 g per litre**
Chlorine for HLD by soak	**0.1%** (JIK 1 : 34) for 20 min in general IP manuals; some MVA guidance uses **0.5% for 20 min**. Follow the unit protocol
HLD by boiling or steaming	**20 minutes** (timed from the rolling boil)
Glutaraldehyde 2%	**20 min = HLD**; **8–10 h = sterilisation**
Autoclave	**121 °C, 106 kPa (15 lb/in²), 20 min unwrapped / 30 min wrapped**
Dry heat	**170 °C for 60 min** (or 160 °C for 120 min)
HLD items storage	Covered HLD container, up to about **1 week**
Safety box	Close at **three-quarters full**
Blood spill	Cover with **0.5% chlorine for about 10 min**
HAI definition	Appears **≥48 h** after admission, or within **30 days** of surgery (**90 days** with an implant)
Droplets vs airborne	Droplets **>5 µm**, travel about **1–2 m**; airborne **<5 µm** (TB, measles, varicella) need an **N95**

Item	Value
CS antibiotic prophylaxis	Within **60 min before the skin incision**
VE frequency in active first stage	About **every 4 hours** unless indicated
HIV PEP	**TLD (TDF 300 / 3TC 300 / DTG 50) 1 tablet daily for 28 days**; start ideally within **2 h**, never after **72 h**
PEP follow-up	Review at **2 weeks**; HIV test at about **6 weeks and 3 months**
Transmission risk after needle-stick	**HBV up to 30%** (HBeAg+), **HCV about 1.8%**, **HIV about 0.3%** (mucous membrane about 0.09%)
HBIG	As soon as possible, ideally within **24 h** (up to 7 days)
Protective anti-HBs	**≥10 mIU/mL**
HBV vaccine schedule	**0, 1 and 6 months**
VHF contacts	Monitor for **21 days**

Exam traps

- **"Gloves replace hand hygiene."** No. Clean hands before (if a moment applies) and **always after** removing gloves.
- **"Wait for the source's HIV result before starting PEP."** No. **Start PEP now**; stop it if the source is negative.
- **"Squeeze the wound and pour bleach or spirit on it."** No. Let it **bleed freely**, wash with **soap and running water**.
- **"PEP can still be started on day 4."** No. Beyond **72 hours** PEP is not recommended; test and follow up instead.
- **"An HIV-positive health worker gets PEP."** No. She needs full **ART** and linkage to care.
- **"Leave instruments in chlorine overnight to be extra safe."** No. **10 minutes** only; longer corrodes the steel.
- **"Just sterilise the dirty instruments."** No. **Clean first**: you cannot sterilise dirt.
- **"Use alcohol rub on visibly soiled hands."** No. Use **soap and water** when hands are visibly soiled.

Questions seniors ask

Viva questions

1. **List the components of standard precautions.** Hand hygiene, PPE by risk, respiratory hygiene and cough etiquette, safe injections and sharps, environmental cleaning, linen and waste management, reprocessing of equipment, and patient placement.
2. **Name the 5 moments.** Before touching a patient; before a clean/aseptic procedure; after body-fluid exposure risk; after touching a patient; after touching patient surroundings.
3. **Why decontaminate instruments before cleaning?** To inactivate HIV, HBV and HCV so that the person cleaning them is protected.
4. **Autoclave parameters?** 121 °C at 106 kPa (15 lb/in²), 20 minutes unwrapped and 30 minutes wrapped.
5. **How do you prepare 0.5% chlorine from 3.5% JIK?** 1 part JIK to 6 parts water.
6. **Risk of HIV transmission after a needle-stick from an HIV-positive source?** About 0.3% (mucous membrane about 0.09%), and much lower if the source is virally suppressed or PEP is taken.
7. **PEP regimen, timing and duration?** TDF/3TC/DTG once daily for 28 days, started as soon as possible (ideally within 2 h, no later than 72 h).
8. **Which colour bin for a blood-soaked swab? For an expired drug?** Yellow (infectious). Brown (pharmaceutical).

9. **When must you use soap and water instead of alcohol rub?** When hands are visibly soiled or contaminated with body fluids, after using the toilet, and with spore-forming organisms (e.g. *C. difficile*) or during diarrhoeal outbreaks.

On the ward round

1. **You have just had a needle-stick. What do you do, in order?** Make the patient safe; let it **bleed**, **wash with soap and water**; **report**; assess the exposure and source; baseline HIV test; **TLD within 2 h (max 72 h) for 28 days**; hepatitis B plan; counselling; follow-up tests at 6 weeks and 3 months.
2. **Which exposures do NOT need PEP? Intact skin** contact, and contact with urine, faeces, saliva or vomit **without visible blood**; and any exposure more than **72 hours** earlier.
3. **The source is HIV-positive with an undetectable viral load last month. Do you still give PEP?** The risk is very low, but most Ugandan protocols still give PEP when there is **any doubt** (an old result, poor adherence). Discuss with the PEP provider; don't simply refuse it.
4. **How do you recognise an outbreak? More cases than expected** in a particular place and time, often with the **same organism**. Notify the IPC focal person, write a **case definition**, make a **line list**, audit practices, control, and monitor.
5. **Why are routine environmental swabs not recommended?** They are expensive and hard to interpret. Take them **only when the epidemiology points to a source**.
6. **What is the difference between decontamination, HLD and sterilisation?** Decontamination (0.5% chlorine 10 min) makes items safe to handle; **HLD** kills all organisms except some spores; **sterilisation** kills everything, including spores.
7. **A health worker had 3 HBV vaccine doses but her anti-HBs is 4 mIU/mL; the source is HBsAg-positive. What do you give?** She is a **non-responder**: give **HBIG** and **revaccinate** (or HBIG twice), as in the hepatitis B table.

Mnemonics

- **5 Moments = "2 Befores, 3 Afters"**: Before patient, Before aseptic; After fluids, After patient, After surroundings.
- **Instrument processing "D-C-H/S-S"**: **D**econtaminate (0.5% chlorine, 10 min), **C**lean, **H**LD or **S**terilise, **S**tore.
- **Autoclave "121-106-20-30"**: 121 °C, 106 kPa, 20 min unwrapped, 30 min wrapped.
- **HLD "Boil 20 from the bubble"**: start the clock at the rolling boil.
- **JIK "1 + 6 = 0.5"**: 1 part 3.5% JIK plus 6 parts water gives 0.5% chlorine.
- **Spaulding "Sterile tissue gets Sterilised; Mucosa gets HLD; Skin gets cleaned."**
- **Donning "Gown, Mask, Goggles, Gloves: G-M-G-G"**; doffing **reverses the dirtiest first: Gloves, Goggles, Gown, Mask.**
- **Needle-stick "BLEED-WASH-REPORT-PEP"** within **2 hours ideally, 72 hours maximum**.
- **Risk "Rule of 3s"**: HBV **30%**, HCV **3%** (actually about 1.8%, so remember it as "under 3"), HIV **0.3%**.
- **Bin colours "Black is Benign, Yellow is Yucky, Red is Really dangerous, Brown is for Bottles of drugs."**

OSCE tips

> **EXAM FAVOURITE**
> The UCU practical sessions (Sr Kiggundu, Dr Kaduyu) typically test **hand rub technique, donning and doffing, chlorine preparation, instrument processing sequence and waste segregation**. Practise saying the steps **out loud as you do them**. Examiners mark what they can see and hear.

- State the moment before you start hand rub, and keep rubbing until dry.
- In doffing, gloves come off first and hand hygiene comes last.

- For the chlorine station, show the formula as well as the answer.

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SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	What are standard precautions?	The minimum IPC practices for **every patient, every time**: hand hygiene; PPE by risk assessment; respiratory hygiene; safe injections and sharps; environmental cleaning; linen and waste management; reprocessing of equipment; patient placement.
2	List the WHO 5 Moments for hand hygiene.	1. **Before** touching a patient 2. **Before** a clean/aseptic procedure 3. **After** body-fluid exposure risk 4. **After** touching a patient 5. **After** touching patient surroundings
3	How long for an alcohol hand rub vs a soap-and-water handwash? When must you use soap and water?	Rub: **20–30 s**. Wash: **40–60 s**. Use soap and water when hands are **visibly soiled** or contaminated with body fluids, after the toilet, and with **spore-formers** (C. difficile) or in diarrhoeal outbreaks.
4	Sequence of instrument processing.	1. **Decontaminate**: 0.5% chlorine for **10 min** 2. **Clean**: detergent, water, brush, under water; rinse; dry 3. **HLD** (boil 20 min) **or sterilise** (autoclave) 4. **Store** clean, dry and covered
5	Autoclave parameters.	**121 °C, 106 kPa (15 lb/in²)**: **20 min unwrapped, 30 min wrapped**. Use chemical indicator tape on every pack and record each cycle.
6	Three methods of high-level disinfection and their times.	**Boiling 20 min** (timed from the rolling boil); **steaming 20 min**; **chemical soak 20 min** in 0.1% chlorine or 2% glutaraldehyde, then rinse with boiled water.
7	Difference between HLD and sterilisation.	**HLD** kills all microorganisms **except some bacterial spores**. **Sterilisation** kills **all**, including spores.
8	Spaulding classification with O&G examples.	**Critical** (sterile tissue: CS instruments, curettes) need **sterilisation**. **Semi-critical** (mucous membranes: specula, MVA cannulae) need at least **HLD**. **Non-critical** (intact skin: BP cuff, Pinard) need **cleaning** ± low-level disinfection.
9	How do you prepare 0.5% chlorine from JIK (3.5%)? What is the general formula?	**1 part JIK + 6 parts water.** Total parts water = (% concentrate ÷ % desired) − 1 = (3.5 ÷ 0.5) − 1 = 6. For HTH powder (70%), use about **7 g per litre**. Make it fresh daily.
10	Uganda health-care waste colour codes.	**Black**: non-infectious general waste. **Green**: non-infectious plastics. **Yellow**: infectious waste. **Red**: highly infectious waste. **Brown**: pharmaceutical/chemical waste. **Yellow safety box**: sharps (close at three-quarters full).
11	Order of donning and doffing PPE.	**Donning**: hand hygiene, then gown, mask/respirator, goggles/face shield, gloves (over the cuffs). **Doffing**: gloves, hand hygiene, goggles/face shield, gown (inside out), mask (by the ties), hand hygiene.
12	Needle-stick first aid.	Let it **bleed freely**. **Do not squeeze or suck. Wash with soap and running water.** No bleach or alcohol in the wound. For eyes or mouth, **flush with water or saline for 10–15 min**. Then **report immediately**.

#	Question	Answer
13	HIV PEP regimen, timing and duration (Uganda/WHO).	**TDF 300 mg / 3TC 300 mg / DTG 50 mg (TLD), 1 tablet once daily for 28 days.** Start **as soon as possible**, ideally within **2 h**, **no later than 72 h**. Give the full 28-day supply at the start.
14	Occupational transmission risk after a percutaneous injury from an infected source.	**HBV up to 30%** (HBeAg+), **HCV about 1.8%**, **HIV about 0.3%** (mucous membrane about 0.09%). HBV is the biggest risk, and it is vaccine-preventable.
15	Hepatitis B management after exposure in an unvaccinated health worker (source HBsAg-positive).	**HBV vaccine dose 1 now** (complete 0, 1, 6 months) **plus HBIG** as soon as possible, ideally within 24 h (up to 7 days), where available.
16	Airborne vs droplet precautions.	**Airborne** (TB, measles, varicella): **N95** respirator for staff, a masked patient, a ventilated or single room. **Droplet** (influenza, COVID-19, meningococcus): a **medical mask** within 1–2 m, patient masked, spacing.
17	How do you manage a blood spill on the labour ward floor?	Utility gloves (plus apron and eye protection if large). **Cover with 0.5% chlorine** for about **10 min**, wipe with disposable cloths into the **yellow bin**, clean with detergent and water, then dry.
18	Five IPC practices specific to the labour ward.	1. Hand hygiene and **sterile gloves before every VE**; VEs every 4 h 2. **Sterile delivery set** for each woman 3. **Full PPE incl. eye protection** at delivery 4. **Chlorhexidine 7.1% cord care** and tetracycline eye ointment 5. Clean and disinfect the bed and mackintosh after each birth; one woman per bed
19	Five elements of the WHO SSI prevention bundle.	**No shaving** (clip); **antibiotic prophylaxis before incision**; **surgical hand antisepsis**; **chlorhexidine-alcohol skin prep**; **no prolonged post-op antibiotics** (plus normothermia, glucose control, gentle handling).
20	Follow-up after HIV PEP.	Review at about 2 weeks (adherence, side effects). **Repeat HIV test at about 6 weeks and 3 months** (some protocols add 6 months). Use condoms and don't donate blood until the 3-month test is negative. Follow up HBV/HCV per exposure.
21	Which exposures are significant enough to need an HIV PEP assessment, and which are not?	**Significant**: **percutaneous** (needle-stick, cut), **mucous membrane** (eye, mouth) or **non-intact skin** exposure to **blood, bloody fluid, liquor, vaginal secretions**, semen or CSF. **Not significant (no PEP)**: **intact skin**, or urine, faeces, saliva, vomit **without visible blood**; or an exposure more than **72 h** ago (test and follow up instead).
22	Baseline tests after a needle-stick: for the worker and for the source.	**Worker**: rapid **HIV test** (positive = ART, not PEP), **HBsAg** and **anti-HBs** (≥10 mIU/mL = protected), HCV antibody where available, **pregnancy test** if relevant, creatinine if available (**not needed before the first PEP dose**). **Source (with consent)**: rapid **HIV test**, **viral load** from records, **HBsAg**, HCV antibody. **Start PEP without waiting** for the results.

#	Question	Answer
23	Steps of investigating a suspected cluster of post-CS surgical site infections.	1. **Recognise**: more cases than expected in a place and time 2. **Notify** the IPC focal person and ward leadership 3. **Treat** patients (drain, culture-guided antibiotics); **contact precautions** 4. **Case definition** and **line list** (surgery date, theatre, surgeon, antibiotic timing, onset day, culture) 5. **Audit practice**: prophylaxis timing, skin prep, autoclave log and indicators, chlorine, hand rub, multi-dose vials 6. **Control** the breach, then **monitor and feed back** (SSI surveillance to 30 days)
24	What do you look at on an IPC 'walk-through' of a ward during an outbreak?	**Alcohol rub at the point of care** and working sinks with soap; **chlorine buckets made today**; **autoclave logbook and indicator tape**; sterile packs dry, intact, in date; **safety boxes** within reach and ≤¾ full; **colour-coded waste** used correctly; **multi-dose vials** labelled and dated; overcrowding (two per bed); shared equipment; and staff **hand hygiene at the 5 moments** (observation audit).
25	When are environmental swabs and organism typing useful in an outbreak?	**Environmental swabs**: only when the **epidemiology points to a specific source** (e.g. a shared suction machine or IV fluid). Routine swabbing is **not** recommended. **Typing** (national reference labs, rarely available) can confirm that cases share the same strain, but **you do not need it to start control measures**.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. An intern at Kawempe sustains a deep needle-stick with a hollow-bore needle after drawing blood from a woman with HIV who defaulted on ART 6 months ago. It happened 1 hour ago. He has washed the wound. His last HIV test was negative a year ago. What is the MOST appropriate management now?

A Test himself for HIV at 3 months and start ART only if positive
B Wait for the source's viral load result before deciding on PEP
C Start zidovudine monotherapy for 28 days
D Squeeze the wound and apply undiluted bleach
E Baseline HIV test and start TDF/3TC/DTG once daily for 28 days immediately

2. A midwife finishes an abdominal palpation on a woman in labour. She is about to put on sterile gloves to perform a vaginal examination. According to the WHO 5 Moments, what should she do?

A Nothing, because the sterile gloves will protect the patient
B Perform hand hygiene immediately before putting on the sterile gloves (Moment 2)
C Perform hand hygiene only after the VE (Moment 4)
D Perform hand hygiene only if her hands are visibly dirty
E Wipe her hands on a clean towel and proceed

3. An HC III has no autoclave. After an MVA, the in-charge asks a student how to process the metal vaginal speculum for reuse. What is the CORRECT sequence?

A Soak in 0.5% chlorine overnight, then reuse
B Wipe with methylated spirit and store in a drawer
C Soak in 0.5% chlorine for 10 minutes, scrub with detergent and water, rinse, then boil for 20 minutes from the rolling boil
D Boil for 5 minutes without prior cleaning
E Wash with soap and water only, as it does not enter sterile tissue

4. A ward has run out of ready-made chlorine solution. The only product available is JIK bleach labelled 3.5% available chlorine. How should the student prepare 0.5% chlorine for decontaminating instruments?

A 1 part JIK to 6 parts water
B 1 part JIK to 1 part water
C 1 part JIK to 9 parts water
D 1 part JIK to 34 parts water
E Use the JIK undiluted

5. After conducting a delivery, a student has a blood-soaked gauze swab, a used suture needle, the empty oxytocin ampoule, and the paper wrapper of the sterile gloves. Which disposal is CORRECT under Uganda's colour coding?

A All items in the yellow bin
B Gauze in the black bin; needle and ampoule in the yellow bin; wrapper in the red bin
C Needle recapped and placed in the yellow bin; the rest in the black bin
D Gauze in the yellow bin; needle and broken ampoule in the safety box; wrapper in the black bin
E Gauze in the brown bin; needle in the black bin; wrapper in the yellow bin

6. A student is about to conduct a normal delivery for a woman of unknown HIV status who arrived in the second stage. Which PPE is MOST appropriate?

A Examination gloves only
B Sterile gloves and a surgical cap
C Sterile gloves, plastic apron, mask and eye protection (goggles or face shield), and closed shoes/boots
D Full Ebola coverall with double gloves and a hood
E No PPE until her HIV test result is available

7. A nurse who never completed hepatitis B vaccination sustains a needle-stick from a patient who is HBsAg-positive. She is also HIV-negative. The source is HIV-negative. What is the MOST appropriate hepatitis B management? Advanced

A No action, since the source is HIV-negative
B Test for anti-HBs at 6 months and vaccinate if negative
C Tenofovir alone for 28 days
D Interferon injections for 3 months
E Hepatitis B vaccine dose now (completing 0, 1 and 6 months) plus hepatitis B immunoglobulin as soon as possible

8. After caring for a woman on contact and droplet precautions, a student wearing gloves, gown, mask and face shield is about to remove the PPE. Which doffing sequence is CORRECT?

A Gloves, hand hygiene, face shield, gown, mask, hand hygiene
B Mask, face shield, gown, gloves, hand hygiene
C Face shield, mask, gloves, gown, hand hygiene
D Gown, mask, face shield, gloves, no hand hygiene needed if gloves were worn
E Mask, gloves, gown, face shield, hand hygiene

9. In two weeks, 6 of 40 women (15%) who had CS in a regional referral hospital's theatre develop wound infections with similar organisms. The usual SSI rate is about 5%. As the intern on the IPC committee, what is the BEST next step? Advanced

A Prescribe 7 days of ceftriaxone to every woman after CS from now on
B Notify the IPC focal person, treat cases with culture-guided antibiotics, and investigate practices (antibiotic timing, skin prep, autoclave logs and indicators, theatre cleaning, hand-rub availability)
C Close the theatre permanently
D Suspend the surgeon who operated on most of the women
E No action, since SSI is common after CS in Uganda

10. On the postnatal ward, a 27-year-old HIV-positive woman has had a cough for 3 weeks, with night sweats and weight loss. She shares a room with 5 mothers and their newborns. What is the MOST appropriate IPC action?

A Droplet precautions: staff wear a surgical mask only when within 1 metre
B Contact precautions: gloves and gown for touching her
C No precautions until the sputum result is available
D Airborne precautions: give her a surgical mask, move her to a well-ventilated single room, staff wear N95 respirators, and send sputum for GeneXpert
E Discharge her home immediately to protect other patients

11. A student opens a CS pack from the store and notices that the chemical indicator tape has NOT changed colour, although the pack is dry and intact. What should be done?

A Do not use it; treat it as unsterile, reprocess it, and report a possible autoclave failure
B Use it, since the pack is dry and intact
C Use it after spraying the instruments with alcohol
D Boil the instruments for 5 minutes and then use them
E Use it for the skin incision but not for the uterus

12. While conducting a delivery, a student's glove tears and her hand becomes visibly contaminated with blood. There is no cut on her skin. What is the MOST appropriate hand hygiene?

A Alcohol hand rub for 20–30 seconds
B Wipe off the blood with gauze and put on new gloves
C Wash with soap and running water for 40–60 seconds, dry with a single-use towel, then re-glove
D Apply povidone-iodine skin prep
E Put on a new glove over the torn one

13. A health worker comes to the clinic 80 hours after a needle-stick from an HIV-positive source because she was afraid to report it. Her baseline HIV test is negative. What is the MOST appropriate management? Advanced

A Start TDF/3TC/DTG now for 28 days
B Start a double dose of TLD to compensate for the delay
C Start TLD for 56 days instead of 28
D Reassure her that no further action is needed
E PEP is not recommended beyond 72 hours. Counsel her, offer follow-up HIV testing at 6 weeks and 3 months, check HBV status and vaccinate, and address the barriers to reporting

14. Immediately after a delivery, there is a large pool of blood and liquor on the labour ward floor. What is the CORRECT way to manage the spill?

A Mop it up immediately with water and the ward mop
B Wear utility gloves, cover the spill with 0.5% chlorine for about 10 minutes, wipe up with disposable cloths into the yellow bin, then clean with detergent and water
C Sprinkle undiluted JIK on it and leave it to dry
D Cover it with sawdust and sweep it into the black bin
E Leave it for the cleaners at the end of the shift

15. A 22-year-old primigravida has an uncomplicated spontaneous vaginal delivery with a mediolateral episiotomy, repaired without complications. A student suggests 5 days of oral amoxicillin 'to prevent infection'. What is the MOST appropriate response?

A Agree; amoxicillin 500 mg 8-hourly for 5 days
B Give ceftriaxone 1 g IV once daily for 3 days
C Give metronidazole 400 mg 8-hourly for 7 days
D No routine antibiotics; emphasise hand hygiene, perineal hygiene and review for signs of infection
E IV ampicillin plus gentamicin for 48 hours

Answers & explanations

1. E · Baseline HIV test and start TDF/3TC/DTG once daily for 28 days immediately

This is a **high-risk exposure** (deep, hollow-bore, blood, source likely viraemic). Start **PEP with TLD as soon as possible**, within 72 h and ideally within 2 h, after a baseline HIV test (**E**). **A** misses the window in which infection can be prevented. **B**: never delay PEP for source results; stop later if the source is negative. **C**: monotherapy is obsolete and less effective. **D**: squeezing and caustics damage tissue and do not reduce transmission.

2. B · Perform hand hygiene immediately before putting on the sterile gloves (Moment 2)

Moment 2 (before a clean/aseptic procedure) applies **even when gloves will be worn**. Hands contaminate the outside of gloves during donning, and gloves have micro-perforations (**B**). **A** is a classic misconception. **C**: hygiene after the VE is Moment 3/4 and does not protect the patient from the VE. **D**: organisms are present on hands that look clean. **E**: wiping does not remove organisms.

3. C · Soak in 0.5% chlorine for 10 minutes, scrub with detergent and water, rinse, then boil for 20 minutes from the rolling boil

A speculum is **semi-critical** (it touches mucous membranes), so it needs at least **HLD**. The correct chain is **decontaminate (0.5% chlorine, 10 min), clean, then HLD (boil 20 min)** (**C**). **A**: prolonged chlorine corrodes instruments, and chlorine alone is not HLD after dirty instruments. **B**: spirit is not HLD. **D**: 5 minutes is too short, and without cleaning, organic material shields microbes. **E**: cleaning alone is for non-critical items.

4. A · 1 part JIK to 6 parts water

Total parts water = (% concentrate ÷ % desired) − 1 = (3.5 ÷ 0.5) − 1 = **6**, so the dilution is **1:6** (**A**). **B** gives about 1.75% (too strong and corrosive). **C** gives 0.35% (too weak). **D** gives **0.1%**, which is the concentration for HLD soaks, not decontamination. **E** is 3.5%: wasteful, corrosive and hazardous.

5. D · Gauze in the yellow bin; needle and broken ampoule in the safety box; wrapper in the black bin

Blood-soaked items are **infectious (yellow)**. Needles and **glass ampoules are sharps (safety box)**. Clean paper is **general non-infectious (black)** (**D**). **A** overloads costly infectious-waste streams and puts sharps in a bag (injuring waste handlers). **B** puts infectious gauze in general waste. **C** involves recapping (a needle-stick risk) and a sharp in a bag. **E** is wrong for all items: brown is for pharmaceuticals.

6. C · Sterile gloves, plastic apron, mask and eye protection (goggles or face shield), and closed shoes/boots

Delivery carries a high **splash risk** (liquor, blood) to the face, eyes, body and feet. Standard precautions call for **sterile gloves, an apron, a mask and eye protection, and closed footwear** (**C**). **A** and **B** leave the mucous membranes and clothing exposed. **D** is for suspected VHF, not routine care. **E** is wrong: standard precautions apply to everyone regardless of status, and waiting delays care.

7. E · Hepatitis B vaccine dose now (completing 0, 1 and 6 months) plus hepatitis B immunoglobulin as soon as possible

An unvaccinated worker exposed to an HBsAg-positive source should receive **HBV vaccine plus HBIG** as soon as possible (ideally within 24 h, up to 7 days), then complete the schedule (**E**). **A**: HBV (not HIV) is the risk here, and it is the most transmissible blood-borne virus. **B**: waiting 6 months misses prevention. **C**: tenofovir has HBV activity but is not standard HBV PEP. **D**: interferon treats chronic infection; it is not prophylaxis.

8. A · Gloves, hand hygiene, face shield, gown, mask, hand hygiene

Remove the **most contaminated item first (gloves)**, clean the hands, then remove the face shield, the gown (inside out), and finally the **mask by its ties**, followed by hand hygiene (**A**). **B**, **C** and **E** remove the mask or face shield with contaminated gloved hands, touching the face area. **D** omits hand hygiene; gloves do not replace it.

9. B · Notify the IPC focal person, treat cases with culture-guided antibiotics, and investigate practices (antibiotic timing, skin prep, autoclave logs and indicators, theatre cleaning, hand-rub availability)

A cluster above the baseline rate is a **suspected outbreak**. Notify, treat, and **investigate systematically** to find and fix the breach (**B**). **A** worsens resistance (the WHO recommends against prolonged prophylaxis) and ignores the cause. **C** is disproportionate and harms access; a temporary closure for cleaning may be part of the plan. **D**: blame is premature and a system failure is more likely (e.g. autoclave failure). **E** ignores a signal of preventable harm.

10. D · Airborne precautions: give her a surgical mask, move her to a well-ventilated single room, staff wear N95 respirators, and send sputum for GeneXpert

Suspected **pulmonary TB** is spread by **airborne** droplet nuclei. Mask the patient (source control), **separate** her in a ventilated room, give staff **N95** respirators, and **investigate urgently** (GeneXpert) (**D**). **A** is inadequate for airborne spread. **B** does not address respiratory transmission. **C** exposes newborns and staff while waiting. **E** abandons her diagnosis and care and does not protect her baby.

11. A · Do not use it; treat it as unsterile, reprocess it, and report a possible autoclave failure

An unchanged indicator means the pack **may not have reached sterilising conditions**. It must be treated as **unsterile**, reprocessed, and the autoclave checked and the problem reported (other packs from that cycle are suspect too) (**A**). **B**: dry and intact does not prove sterility. **C** and **D**: alcohol and 5-minute boiling are not sterilisation. **E**: all CS instruments are critical items.

12. C · Wash with soap and running water for 40–60 seconds, dry with a single-use towel, then re-glove

Visibly soiled hands need **soap and running water** (alcohol does not remove organic material well) (**C**). Intact skin exposure needs no PEP, but the incident should still be noted. **A** is for hands that are not visibly soiled. **B** leaves contamination. **D**: skin prep is not hand hygiene. **E** traps blood against the skin.

13. E · PEP is not recommended beyond 72 hours. Counsel her, offer follow-up HIV testing at 6 weeks and 3 months, check HBV status and vaccinate, and address the barriers to reporting

PEP is **not recommended after 72 hours**, because by then it is unlikely to prevent infection (**E**). Follow-up testing still matters, so that any infection is caught early and she starts ART. Also manage HBV and address the fear of reporting. **A**, **B** and **C** give drugs outside the effective window (and doubling or extending doses is not evidence-based). **D** neglects follow-up and HBV.

14. B · Wear utility gloves, cover the spill with 0.5% chlorine for about 10 minutes, wipe up with disposable cloths into the yellow bin, then clean with detergent and water

Spills are **inactivated with 0.5% chlorine** (about 10 min contact), removed as **infectious waste (yellow)**, and the area is then cleaned with detergent (**B**). **A** spreads contamination over the floor and mop. **C**: undiluted bleach is hazardous and not needed. **D**: it puts infectious material in the general waste. **E** leaves a slip and infection hazard.

15. D · No routine antibiotics; emphasise hand hygiene, perineal hygiene and review for signs of infection

The WHO recommends **against routine antibiotic prophylaxis** for uncomplicated vaginal birth and for episiotomy. It gives no proven benefit and drives antimicrobial resistance, which is already severe in Uganda (**D**). Antibiotics **are** indicated for manual removal of the placenta, 3rd/4th-degree tears and CS. **A**, **B**, **C** and **E** are unnecessary antibiotic use.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Needle-stick during a night shift

At 02:00 on the labour ward at Mengo, you (a Year 3 student) are assisting with the repair of a second-degree perineal tear. The woman is 26 years old and HIV-positive on TLD; her last viral load, 8 months ago, was 'detectable, 1,200 copies/mL'. The suture needle slips and pierces your glove and finger, drawing blood.

1. What do you do in the first 5 minutes?

1. Tell the midwife or doctor and step away safely (the repair can be completed by someone else).
2. Remove the gloves and **let the wound bleed freely**. **Do not squeeze or suck.**
3. **Wash with soap and running water** for several minutes. No bleach or alcohol.
4. Cover it with a waterproof dressing.
5. **Report immediately** to the in-charge (on call) and fill in the incident form.

2. How do you assess the risk?

- **Exposure**: percutaneous, through a glove, with a **solid** suture needle and visible bleeding. This is significant, though lower risk than a hollow-bore needle.
- **Source**: HIV-positive with a **detectable viral load (1,200 copies/mL)**, so not suppressed. That means real risk. Check her HBsAg (and HCV) with consent.
- **You**: baseline HIV status, HBV vaccination history (3 doses?) and anti-HBs, and pregnancy status.

The decision is **PEP indicated**.

3. What tests and treatment do you receive, and when?

- **Baseline rapid HIV test** (with counselling), HBsAg/anti-HBs, and a pregnancy test if relevant. Creatinine if available (don't delay the first dose).
- **Start TDF/3TC/DTG, 1 tablet once daily for 28 days, now**: ideally within 2 hours of exposure, and certainly before 72 h. Take the **full 28-day supply**.
- **HBV**: if fully vaccinated with anti-HBs ≥10, nothing more. If unvaccinated or a non-responder and the source is HBsAg-positive, give the **HBV vaccine plus HBIG**.

4. What counselling and follow-up do you need?

- **Adherence**: the same time daily. Take DTG 2 h before or 6 h after iron, calcium or antacids.
- **Side effects**: nausea, headache and insomnia are usually mild. Report a rash or jaundice.
- **Condoms** and no blood donation until the 3-month test is negative.
- Review at **2 weeks**; **HIV test at about 6 weeks and 3 months**; HCV follow-up if the source is HCV-positive.
- Psychological support. Needle-stick is frightening, and anxiety is normal.

5. How could this injury have been prevented?

- Use **instruments, not fingers**, to hold tissue and handle the needle. Never guide the needle with a finger.
- **Double gloving** for repairs.
- **Good lighting** and adequate analgesia so the woman does not move suddenly.
- A **neutral zone** for the needle and immediate disposal into a safety box.
- Fatigue management at night.
- **Hepatitis B vaccination** for all students before clinical rotations.

Take-away: A needle-stick is an emergency. Bleed, wash and report within minutes, and start TLD within hours (never after 72 h). Always check hepatitis B protection, and fix the practice that caused the injury.

Case 2: Processing a delivery set at an HC III with no autoclave

You are on a community placement at an HC III in Wakiso. After a normal delivery, the enrolled midwife hands you the used delivery set (2 artery forceps, cord scissors, episiotomy scissors, needle holder, toothed forceps, a kidney dish and a speculum). The facility has no autoclave, a charcoal stove, a bottle of JIK (3.5%) and a bucket.

1. What PPE do you need, and what is your first step with the instruments?

Utility gloves, plastic apron and eye protection (splash risk). **Decontaminate**: open all the instruments and fully submerge them in **0.5% chlorine** for **10 minutes** immediately after use (inactivates HIV and HBV). Don't leave them longer, because it corrodes the metal.

2. How do you make the 0.5% chlorine from the JIK?

Total parts water = (3.5 ÷ 0.5) − 1 = **6**. Mix **1 part JIK with 6 parts clean water** (e.g. 1 cup JIK plus 6 cups water). Make it **fresh daily**, keep it covered, and never mix it with acids or other cleaners.

3. Describe the cleaning step and why it is essential.

After decontamination: scrub each instrument **under water** with a soft brush and **detergent**, opening the joints and cleaning the teeth and ratchets. Then **rinse** and **dry**. Cleaning removes blood and tissue that **shield organisms** and inactivate disinfectants. You cannot sterilise or disinfect dirt.

4. How will you disinfect or sterilise these items without an autoclave?

The delivery instruments and speculum are **semi-critical to critical**. Without an autoclave, the accepted fallback is **HLD by boiling**: fully submerge, lid on, **start timing at the rolling boil**, boil for **20 minutes**, and add nothing once timing starts. Remove with HLD forceps, air-dry, and use immediately or store in a covered HLD container (up to about 1 week). The kidney dish is processed similarly. Arrange for items that must be sterile (e.g. for surgical procedures) to be autoclaved at the HC IV.

5. Where does the rest of the delivery waste go?

- **Suture needle, blade, broken oxytocin ampoule**: **safety box** (yellow sharps container), closed at three-quarters full.
- **Blood-soaked gauze, pads, gloves**: **yellow** bin (infectious).
- **Paper wrappers**: **black** bin.
- **Placenta**: give it to the family in a sealed bag if they wish (cultural burial), otherwise the **placenta pit**.
- Clean the delivery couch and mackintosh with detergent, then 0.5% chlorine. Clean any spill with 0.5% chlorine for 10 min first.

Take-away: Decontaminate (0.5% chlorine, 10 min), clean, boil 20 min from the rolling boil, then store. JIK 3.5% is diluted 1:6. Segregate waste at the point of use. Low-resource IPC works if every step is done in order.

Case 3: A cluster of post-caesarean wound infections

At a regional referral hospital, the Ob/Gyn team notices that in the last 3 weeks 9 of 60 women (15%) who had CS have developed wound infections by day 5, compared with the usual 5%. Two have wound dehiscence. Cultures grow ESBL-producing Klebsiella in 5 cases. You are asked to help the IPC focal person.

1. Is this an outbreak, and who needs to know?

Yes. A **cluster above the baseline rate** with a similar organism is a **suspected HAI outbreak**. Notify the **IPC focal person or committee**, the head of department, theatre in-charge, the laboratory and hospital management. Start a **line list** (name, date of CS, surgeon, theatre, time of antibiotic, onset, organism).

2. How should the affected women be managed?

- **Drain** infected wounds (remove sutures) and culture the pus. Daily dressings.
- **Culture-guided antibiotics**: ESBL *Klebsiella* is usually resistant to ceftriaxone, so use a **carbapenem (meropenem)** or another agent per sensitivity, with microbiologist input.
- **Contact precautions**: gloves and apron, dedicated equipment, cohort the women if possible, hand hygiene.
- Return the two women with dehiscence to theatre for mass closure.
- Correct anaemia, check glucose, and give nutrition support.

3. Which practices will you audit to find the source?

- **Antibiotic prophylaxis**: given **before incision** (within 60 min)? Or after cord clamping, or not at all?
- **Skin prep**: which antiseptic, was it allowed to dry, was there shaving on the ward?
- **Surgical hand antisepsis**: soap, water supply, alcohol rub availability.
- **Autoclave**: cycle logs, **indicator tapes**, biological indicators, power cuts, overloaded drums, wet packs.
- **Theatre cleaning** between cases, door traffic, and reuse of single-use items.
- **Multi-dose vials** (e.g. shared lidocaine or oxytocin), IV fluid handling.
- **Post-op wound care**: dressings changed with clean hands? Shared dressing trolleys?
- **Hand hygiene compliance audit** on the postnatal ward.

4. The audit finds the autoclave has been running 15-minute cycles for wrapped packs because of power cuts, and ward staff shave women before CS. What control measures follow?

- **Fix sterilisation**: restore **121 °C at 106 kPa for 30 min for wrapped packs**. Use indicator tapes on every pack, keep a cycle log, and add a generator backup. Recall packs from short cycles and reprocess them.
- **Stop pre-op shaving** (clip only if necessary, in theatre).
- **Reinforce the SSI bundle**: pre-incision antibiotic, chlorhexidine-alcohol prep, the WHO Surgical Safety Checklist Time Out, and no prolonged post-op antibiotics.
- Training and **feedback** to staff; continue **SSI surveillance** to confirm the rate falls back to baseline.
- Report to the regional/MoH IPC structures, and share the lessons without blame.

Take-away: Clusters of infections signal a system breach, often sterilisation, skin prep or antibiotic timing. Treat the patients, then investigate and fix the process. Surveillance and feedback confirm the fix worked.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.